

Presupposition filtering in disjunction: What role does exclusive interpretation play?

Intro. Kalomoiros and Schwarz (2024) found presupposition projection across disjunction is symmetric. This study aims to further investigate whether (and how) **presupposition (Ps)** projection across disjunctions is affected by exclusive disjunctive interpretations.

In terms of theoretical predictions, Mayr and Romoli (2016) observed that Strong Kleene semantics predicts symmetric filtering for inclusive disjunction but no filtering for exclusive disjunction. We here made a novel observation that dynamic semantics also predicts no filtering for exclusive disjunction. (a-h) below show all eight possible ways to define the context change potential of an exclusive disjunction $\alpha \text{ xor } \beta$:

$$\begin{array}{ll} (a) C[\alpha][\neg\beta] \cup C[\beta][\neg\alpha] & (e) C - C[\alpha][\beta] - C[\neg\beta][\neg\alpha] \\ (b) C[\neg\alpha][\beta] \cup C[\neg\beta][\alpha] & (f) C - C[\neg\alpha][\neg\beta] - C[\beta][\alpha] \\ (c) C[\alpha][\neg\beta] \cup C[\neg\alpha][\beta] & (g) C - C[\alpha][\beta] - C[\neg\alpha][\neg\beta] \\ (d) C[\beta][\neg\alpha] \cup C[\neg\beta][\alpha] & (h) C - C[\beta][\alpha] - C[\neg\beta][\neg\alpha] \end{array}$$

Turning to their definedness conditions:

For (a), $C[\alpha]$ and $C[\beta]$ need to be defined, meaning there is no filtering for α and β . For (e), $C[\alpha]$ and $C[\neg\beta]$ need to be defined. Since $C[\neg\beta] = C - C[\beta]$, this means $C[\beta]$ should also be defined, equating the situation with (a). (b) and (f) are also similar to (a).

For (c), $C[\alpha]$, $C[\alpha][\neg\beta]$, and $C[\neg\alpha][\beta]$ should be defined. Since the definedness of $C[\alpha][\neg\beta]$ requires the definedness of $C[\alpha][\beta]$, this means β should be defined on both $C[\alpha]$ and $C[\neg\alpha]$, i.e., the entire context set C . Thus, there is no filtering for both α and β in (c). The cases for (g), (d), and (h) are similar to (c).

Thus, classic trivalent logics and dynamic semantics both predict no filtering for either disjunct in an exclusive disjunction. Assuming exclusivity is gradient, as discovered in Nicolae et al. (2024), this means increasing exclusivity of disjunction reduces filtering.

To test whether exclusivity of disjunction affects its Ps filtering, and if so, whether the above theoretical prediction is borne out, we modified K&S (2024)'s paradigm and conducted a follow-up study in Mandarin (N = 207). We chose Mandarin because it is a sample of convenience and we do not expect significant cross-linguistic differences in this effect. Surprisingly, we found that increasing exclusivity of disjunction enhances filtering.

Design. To manipulate the exclusive interpretation of disjunction, we introduced two cross-subject factors: **disjunction type (DisjType)**: “huozhe” vs. “yaome...yaome...” and environmental **monotonicity (MonoType)**: downward-entailing (DE) vs. upward-entailing (UE)). We expected “yaome yaome” to be more exclusive than “huozhe” and disjunctions in a UE environment to be more exclusive than in DE ones due to exclusive implicatures. These expectations were confirmed in a norming task.

As for the within-subject conditions, we adopted K&S (2024)'s design. They crossed **PresuppositionType (PsType)**: Ps vs. NoPs) with **Order** (First vs. Second), yielding four conditions: PsFirst, PsSecond, NoPsFirst, and NoPsSecond. “PsFirst” means the first disjunct contains the Ps trigger; “NoPsFirst” replaces that Ps trigger in the first disjunct with a suitable non-presuppositional predicate. “PsSecond” and “NoPsSecond” means the Ps trigger (or its NoPs counterpart) is in the second disjunct. To prevent global accommodation, test sentences in these four conditions are all preceded by a context that is **explicitly ignorant (EI)** about the relevant Ps. K&S also included two additional control conditions: EISimplePs, and SSimplePs (the first “S” in “SSimplePs” means the context is **supportive** of the relevant Ps). The difference of ratings between them establishes the cost of local accommodation.

Stimuli. We selected two effective Ps triggers, “jie” (quit) and “zhidao” (know), based on pilot data showing ratings for these in the EISimplePs condition were significantly lower than in SSimplePs. In our main experiment, we created 2 items for each trigger. Each participant completed 15 trials in total: 8 critical trials (2 conditions per item), 2 catch trials, 4 filler

trials, and 1 inferential task. A critical trial consisted of: (i) an either EI or S context; (ii) a test sentence involving a bathroom disjunction; (iii) a judgment task for the naturalness of the test sentence in the context on a 7-point scale.

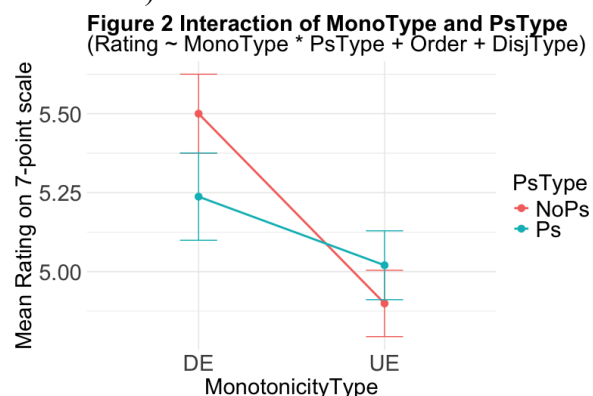
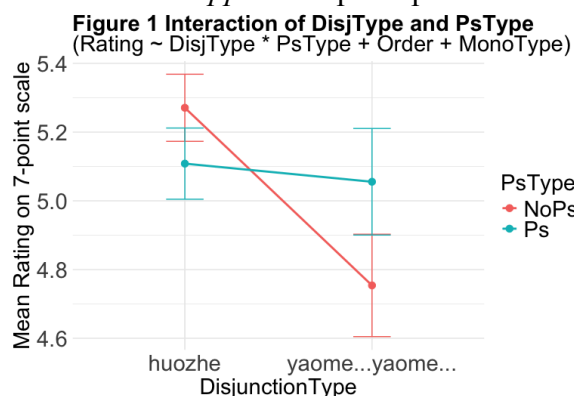
Predictions. (a) If the exclusive interpretation of disjunction affects its Ps filtering, we expect significant two-way interactions for DisjType*PsType and MonoType*PsType.

(b) If that effect is *uniform* for filtering in both disjuncts, we expect no significant three-way interactions including Order (i.e., Order*DisjType*PsType and Order*MonoType*PsType).

(c) If increasing exclusivity *reduces* filtering *uniformly*, we expect (a) and (b) to hold, and that ratings in the Ps conditions should *decrease relative to* the NoPs conditions, when either DisjType changes from “huozhe” to “yaome yaome” or MonoType changes from DE to UE.

Participants. We recruited 207 native Mandarin speakers. 24 of them were excluded for failing at least one of the two catch trials. The remaining 183 responses were analyzed.

Results. We analyzed the data using R’s *clmm* function with a maximal random effects structure. Our analyses revealed significant two-way interactions for DisjType*PsType ($\beta=0.649$, $SE=0.289$, $p=0.025$) and for MonoType*PsType ($\beta=0.611$, $SE=0.290$, $p=0.035$), while neither of the three-way interactions involving Order (Order*DisjType*PsType and Order*MonoType*PsType) reached significance. Unexpectedly, interaction plots (see Fig.1 and Fig.2 below) show that ratings in the Ps conditions actually *increase relative to* the NoPs conditions, when either DisjType changes from “huozhe” to “yaome yaome” or MonoType changes from DE to UE. (Notably, in the “yaome yaome” or UE conditions, ratings in the Ps conditions even exceeded ratings in NoPs conditions. This preference may reflect the *Maximize Presupposition* principle when Ps is felicitous.)



Discussion. Predictions (a) and (b) hold, but (c) does not. Thus, our results suggest increasing exclusivity of disjunction *enhances* filtering uniformly, contrary to the theoretical prediction.

Conclusion. Both trivalent logics and dynamic semantics predict increasing exclusivity reduces filtering uniformly. To test this prediction, we modified K&S (2024)’s experimental paradigm and conducted a follow-up study in Mandarin. Our results indicate that: (1) exclusivity of disjunction does affect its presupposition filtering; (2) this effect is uniform for filtering in both disjuncts; (3) the direction of such effect is opposite to theoretical predictions: increasing exclusivity actually enhances filtering uniformly. This unexpected pattern calls for more empirical validation. If proven to be robust cross-linguistically, it may have profound implications for theories of presupposition projection.

Selected References:

- Kalomoiros, A., & Schwarz, F.** (2024). Presupposition projection from “and” vs “or”: Experimental data and theoretical implications. *Journal of Semantics*, 41(3-4), 331-72.
- Mayr, C., & Romoli, J.** (2016). Satisfied or exhaustified: An ambiguity account of the Proviso Problem. *Proceedings of SALT 26*, 892-912.
- Nicolae, A. C., Petrenco, A., Tsilia, A., & Marty, P.** (2024). Do languages have exclusive disjunctions? *Open Mind*, 8, 1469-1485.